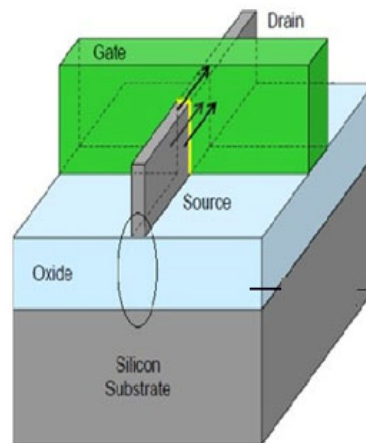


01.119/99.503: Nanoelectronics and Technology

Homework 2 –Brief Solutions

1. Draw the SOI-FinFET structure and describe how current conduction mechanism is different from planar MOSFET. Briefly explain the reason for the improvement of I_{on} and I_{off} characteristics in FinFET compared to planar MOSFET.

Ans:

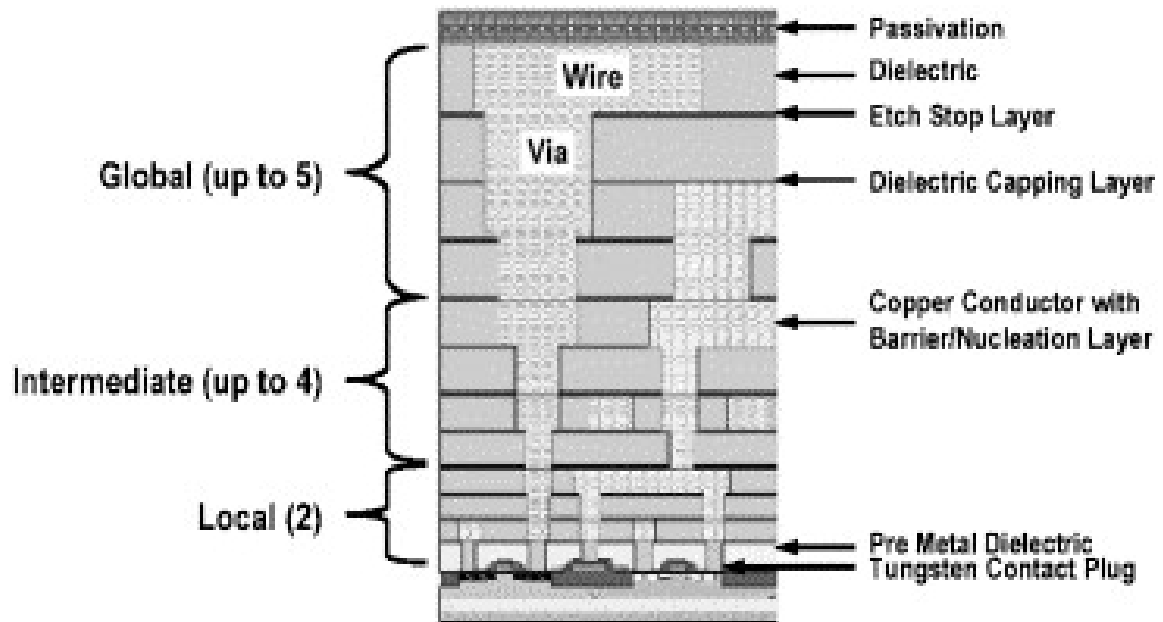


- Un-doped/lightly doped channel- Less scattering
- Better control of V_t and operated at low voltage
- Increased drive (On) current
- No bulk leakage current
- Lower parasitic capacitance
- Requirements of specific photo resist properties

(Refer to reading materials for more explanation)

2. With the help of a block diagram highlight and explain the different interconnect level in CMOS ICs.

Ans:



3. How does interconnect scaling affect the reliability and failure of interconnects.

Ans:

- Electro-migration
- Self-heating
- Stress migration
- Resistivity increases as the feature size decreases
- Increased capacitive coupling and higher line cross talk
- Increase in RC delay than gate delay
- Power (CV^2f) increases
- Reliability issues with interconnect lines

(Refer to study material for explanation on the above concepts)

4. Compare the fundamental deposition mechanisms of Physical Vapor Deposition (PVD), Chemical Vapor Deposition (CVD), and Atomic Layer Deposition (ALD).
→ Highlight differences in source material phase, reaction control, and film growth.
Also include the schematic of the deposition equipments.

Ans:

Physical Vapor Deposition (PVD)

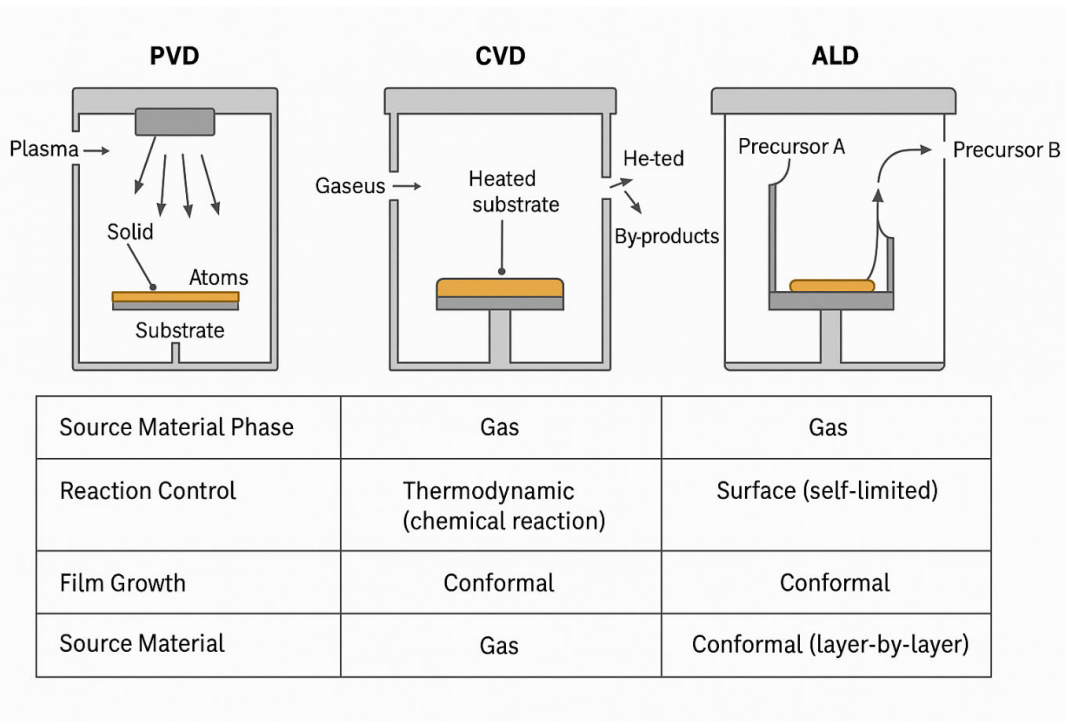
- **Mechanism:** Involves physically moving material from a source to a substrate in a vacuum.
- **Process:** A solid material is vaporized (e.g., through heating or sputtering) and the resulting atoms or molecules travel in a straight line to the substrate to form a film.
- **Key features:**
 - Does not rely on chemical reactions to form the film.
 - Ideal for depositing metals and other non-reactive materials.
 - Often requires lower substrate temperatures compared to CVD.
 - Can have poor step coverage on complex 3D structures.

Chemical Vapor Deposition (CVD)

- **Mechanism:** Uses chemical reactions between gaseous precursors to deposit a solid film on a heated substrate.
- **Process:** Gaseous precursors are introduced into a reaction chamber. They react and/or decompose on the substrate surface, which is typically held at an elevated temperature, to form the desired film.
- **Key features:**
 - Relies on chemical reactions, which can lead to higher deposition rates than ALD.
 - Provides better step coverage than PVD.
 - Can require high temperatures, although plasma-enhanced versions (PECVD) can lower the required substrate temperature.
 - High purity films can be challenging to achieve due to the chemical reactions.

Atomic Layer Deposition (ALD)

- **Mechanism:** Uses a cyclic process of self-limiting surface reactions to deposit the film one atomic layer at a time.
- **Process:** Alternating pulses of different precursor gases are introduced. Each precursor reacts with the surface until the surface is saturated, and a reaction-blocking layer is formed. A purger gas is used to remove any excess reactants before the next pulse is introduced.
- **Key features:**
 - Provides excellent conformity and uniformity on complex 3D structures due to its self-limiting nature.
 - Allows for precise control over film thickness down to a single atomic layer.
 - Slower deposition rates compared to PVD and CVD.
 - Ideal for ultra-thin, precise, and conformal coatings.



5. (a) Briefly explain why polysilicon is not used as a gate electrode material in High-k/Metal Gate (HK/MG) CMOS technology.
- (b) Describe the Gate last process in High-k/Metal Gate (HK/MG) CMOS technology explaining the reason with respect to Gate first process in Poly-Si/SiO₂ CMOS technology.

Ans:

(a)

- Silicide formation.

While the effective capacitance thickness indicated that scaling would eventually require the replacement of SiO₂ by a high K oxide, and of the poly-Si gate by a metal gate, it was expected that the two technological changes could occur separately. However, it became clear that there was a reaction between Si and high K oxide. It was found that the reducing ambient during the CVD deposition poly-Si from silane causes a gross reduction of the ZrO₂ or HfO₂, leading to silicide formation. It was realised that high K gate oxides and metal gates must be introduced simultaneously.

[Ref: J. Robertson and R M Wallace, "High-K materials and Metal Gates for CMOS applications", Materials Science and Engineering: R: Reports, Vol 88 (2015) pg. 1-41]

- Fermi-level pinning and increased threshold voltage.

The second factor was 'Fermi level pinning' which will be described later, and was initially attributed to the formation of Hf-Si bonds at the HfO₂ /poly-Si gate interface. Thus, it was decided to introduce high K oxies and metal gate simultaneously.

[Ref: J. Robertson and R M Wallace, "High-K materials and Metal Gates for CMOS applications", Materials Science and Engineering: R: Reports, Vol 88 (2015) pg. 1-41]

- (b) The metal gates have lower thermal budget. Source/drain process involves high thermal budget. Hence, if gate first process is used, the metal gates will be melted and could diffuse to gate oxide degrading the device.

The gate first process follows the same process flow as with a SiO₂ gate oxide. In gate first, we sequentially deposit a gate oxide layer, gate work function tuning layer, and gate metal. The source and drain doping implants are carried out, and the dopant activation anneal to 1000C is carried out. This annealing temperature is too high for many gate stacks.

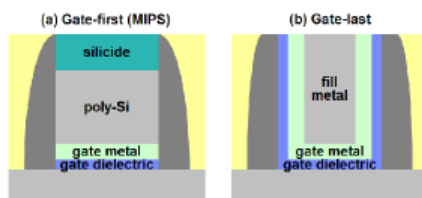
In the gate last process, the gate oxide is deposited followed by a poly-Si dummy gate. The source drain dopant implants are carried out, followed by the dopant activation anneal. An interlayer dielectric is deposited for mechanical stability. This is then planarised by chemical mechanical polishing (CMP) down to the level of the top of the dummy gate. The dummy gate is then etched out, and the metal gate layers are then deposited into this hole, followed by an Al fill. An anneal to 500C is applied. In this way, the gate metal is not exposed to the 1000C temperature anneal.

Variant 2 of the gate-last process etches off both the dummy gate and a 'dummy gate oxide', and replaces both with new gate oxide and gate metal.

Materials Science and Engineering: R: Reports, Vol 88 (2015) pg. 1-41]

High-K /Metal gate CMOS

Gate Last process



- **Reduced Risks of Damage:** High-k dielectrics are sensitive to certain processing steps, especially high temperatures and reactive species. By using the gate last process, the exposure of the high-k material to potentially damaging processes is minimized.
- **Better Interface Quality:** By depositing the gate after the formation of source and drain regions, it's easier to achieve a high-quality interface between the high-k dielectric and the metal gate, which is critical for device performance.
- **Thermal Budget Management:** The gate last approach allows for a lower thermal budget during fabrication. This means that any high-temperature processes, such as activation of dopants in the source and drain regions, occur before the gate is deposited. This minimizes the risk of dopant diffusion into the gate material, which can degrade performance.

[Ref: High-k/metal gate innovations enabling continued CMOS scaling, M. Frank, Published Proceedings of the ESSCIRC \(ESSCIRC\) 2011](#)

6. (a) Explain why diffusion process of doping is less used compared to ion implantation in advanced CMOS technology, even though it is a simpler process.

(b) Compare diffusion and ion implantation in terms of:

- Dopant control precision
- Junction depth
- Process temperature
- Crystal damage
- Equipment complexity

(a) Ans:

Diffusion is less used than ion implantation in advanced CMOS because ion implantation offers far greater precision in controlling the dopant depth and concentration, which is crucial for creating the fine-featured components in modern chips. Ion implantation is also a [lower-temperature process](#), reducing lattice damage, and is more [repeatable](#), making it more reliable for large-scale manufacturing.

Even though diffusion is simpler and cheaper, it lacks the precision, control, and scalability required for nanoscale CMOS technology. Ion implantation, coupled with advanced annealing techniques, ensures shallow, abrupt, and reproducible junctions, aligning perfectly with modern device physics and process integration needs.

(b)

Parameter	Diffusion Doping	Ion Implantation
1. Dopant Control Precision	• Poor control — dopant concentration and depth depend on diffusion coefficient, temperature, and time. • Hard to achieve uniform or abrupt profiles.	• Excellent control — dose and energy precisely controlled by beam current and acceleration voltage. • Allows tailored and reproducible dopant profiles.
2. Junction Depth	• Relatively deep junctions due to long high-temperature exposure. • Difficult to form ultra-shallow junctions needed for modern CMOS.	• Shallow, abrupt junctions achievable by adjusting implant energy and annealing conditions. • Suitable for sub-20 nm devices (FinFET, GAA).
3. Process Temperature	• Very high (~900–1100 °C for extended time). • Large thermal budget can cause unwanted dopant diffusion or layer degradation.	• Implantation at or near room temperature, followed by brief rapid thermal anneal (RTA) at ~1000 °C for milliseconds. • Much lower effective thermal budget.
4. Crystal Damage	• Minimal — diffusion occurs via solid-state atomic motion, preserving lattice quality.	• Causes significant lattice damage (amorphization) due to ion collisions. • Requires post-implant anneal to repair damage and activate dopants.
5. Equipment Complexity	• Simple furnace setup, low cost, easy maintenance. • Limited flexibility for selective doping.	• Complex high-vacuum ion implanter with beam alignment, mass selection, and dose monitoring systems. • High capital and maintenance cost.

7. Using appropriate photolithography steps, obtain the final pattern on the Silicon wafer as shown in Fig. 1, below. Sketch the detailed steps followed during the whole photolithography process and name the type of photolithography used.

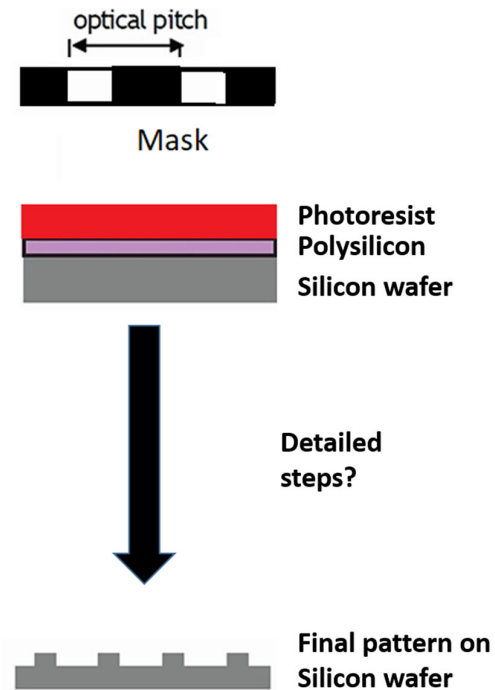


Fig. 1

Ans:

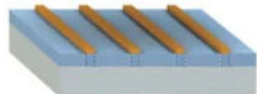
Method: LELE

LITHO-ETCH-LITHO-ETCH (LELE)

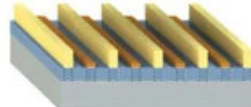
1 Litho 1. The first pattern [yellow] is exposed onto a hard mask.



2 Etch 1. The first pattern is etched into the hard mask [brown].



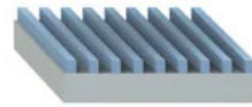
3 Litho 2. A second pattern [yellow] is exposed onto silicon [blue], doubling pattern density.



4 Etch 2. The final, double-density pattern is engraved into the silicon.

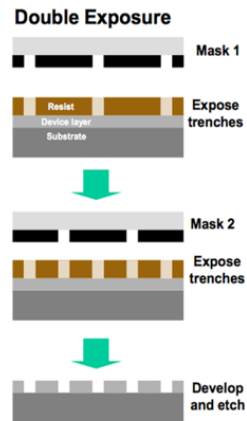


5 Wash. The remaining mask is washed away.

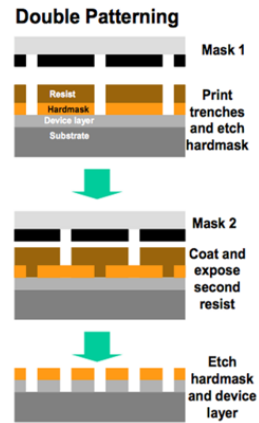


2. SADP: (Also refer to Lecture Slides)

Method 1



Method 2



Ref: <https://www.techdesignforums.com/practice/guides/triple-patterning-self-aligned-double-patterning-sadp/>

Note: Also refer to SADP material discussed in class